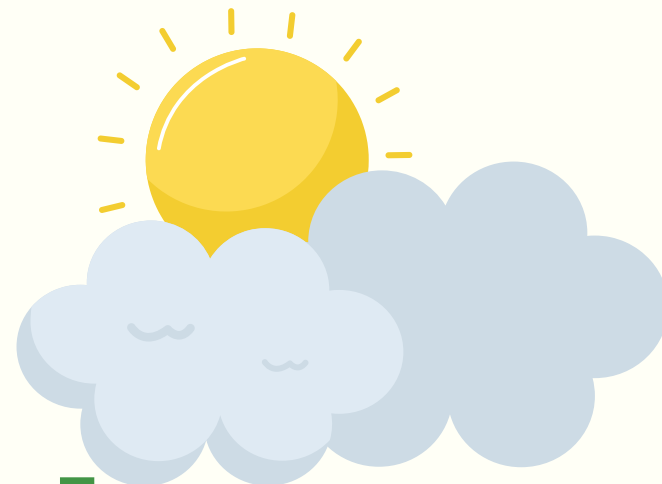
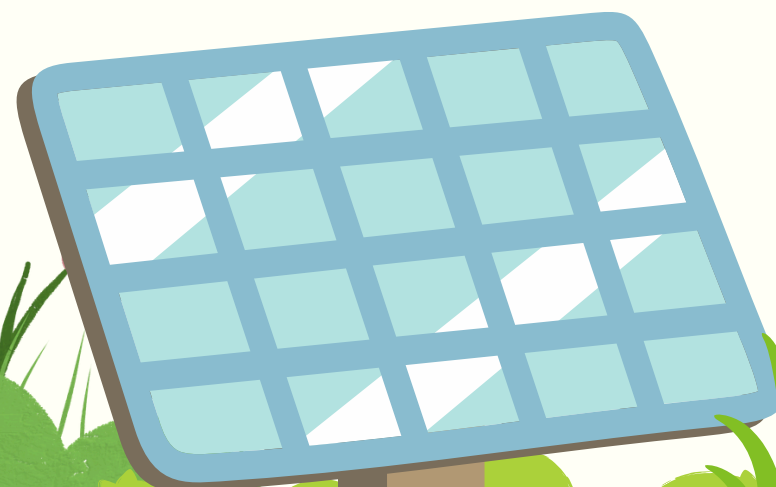




Solar-Battery Micro-Grid Optimizer

**Bridging Renewable Energy with
Automated Peak-Shaving**





Keithston and
Partners

The Global Energy Problem

Why Change Is Needed

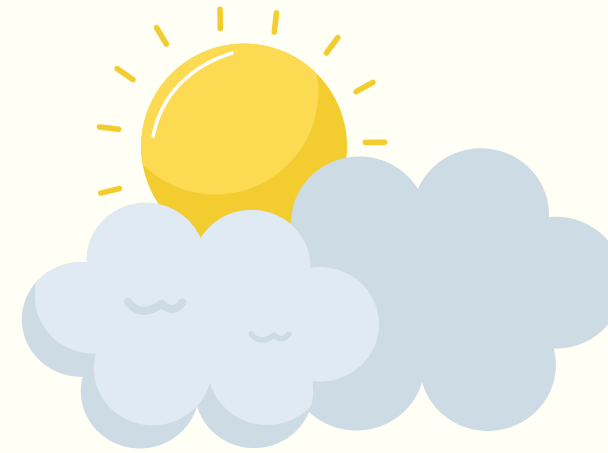
Residential and commercial buildings suffer from high electricity bills and grid strain due to unoptimized power switching during peak-tariff hours

Most sustainability tools rely on passive dashboards ("awareness") rather than active control ("measurement + automation").

Our Mission: Build an autonomous system that actively slashes grid dependency and financial waste.



The Solution: Digital Twin & Optimization Engine



Core Architecture: Built using Python, Pandas, and Streamlit to simulate real-time energy flows.

Autonomous Logic: Automatically detects generation surpluses and peak-tariff windows to route power dynamically.

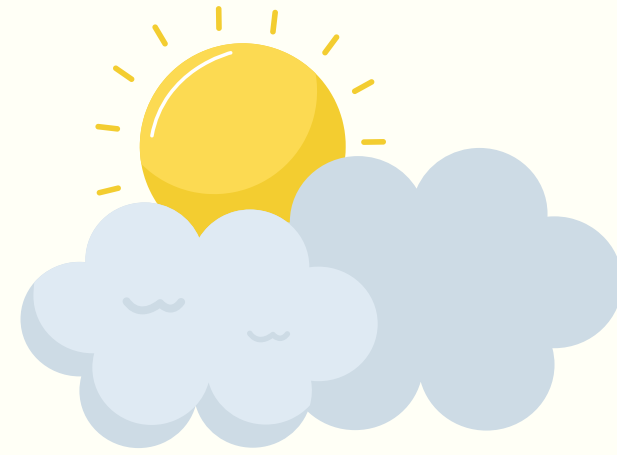
Key Workflows:

High Solar: Excess power is automatically diverted to battery storage.

Peak Tariffs: Expensive grid reliance is bypassed using stored battery power.

Zero Hardware Risk: Fully functional software model designed for instant scalability and zero-failure live demos.





Testing, Validation & Impact Measurement

⚡ kWh avoided

Minimized heavy commercial
grid dependency.

Lowers long-term energy costs

Direct reduction in peak-hour
commercial tariff penalties

🌱 CO2 Reduced

Maximized clean renewable
utilization per cycle.



SCALABILITY

Market Potential: High commercial applicability for offices, residential complexes, and micro-grids facing escalating energy tariffs.

Hardware Integration Ready: Architecture designed to bridge software directly with physical IoT smart meters, solar controllers, and battery management systems (BMS).

Next Steps: Integrating machine learning weather forecasting for 24-hour predictive generation scheduling.



Thank You
So Much!

